Ship/Vessel Emission Measurement and Monitoring

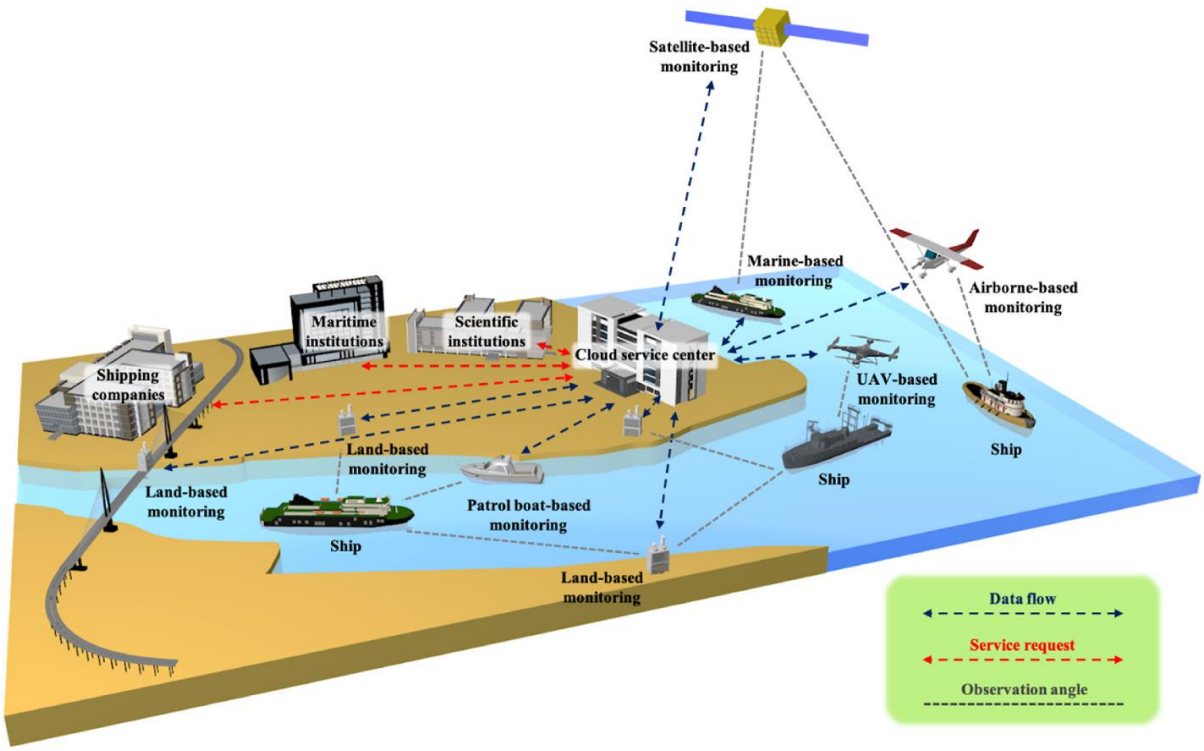
Implementation Plan

1. Establish emission-controlled areas (ECA) –

   An Emission Control Area (ECA) is a sea area in which stricter controls to minimize airborne emissions from ships are established. This regulation is aimed at reducing sulphur oxides (SOX) and is extended in 2005 to include nitrogen oxides (NOX) for several areas.

   The ECA requirements – Ships sailing in emission control areas will have to use fuel oil on board with a sulphur content of no more than 0.10% (mass by mass) [domestic agencies can set its own requirements]. The interpretation of "fuel oil used on board" includes use in main and auxiliary engines and boilers.
   The rules are part of the International Convention for the Prevention of Pollution from ships (MARPOL) Annex VI (Regulations for the Prevention of Air Pollution from Ships), specifically under regulation 13, which cover Nitrogen Oxides (NOx) emissions, and 14, which covers emissions of Sulphur Oxides (SOx) and particulate matter (PM) from ships.

2. Design and construct a ship emission monitoring sensor web (SEMSW)



This will include the following key technologies:

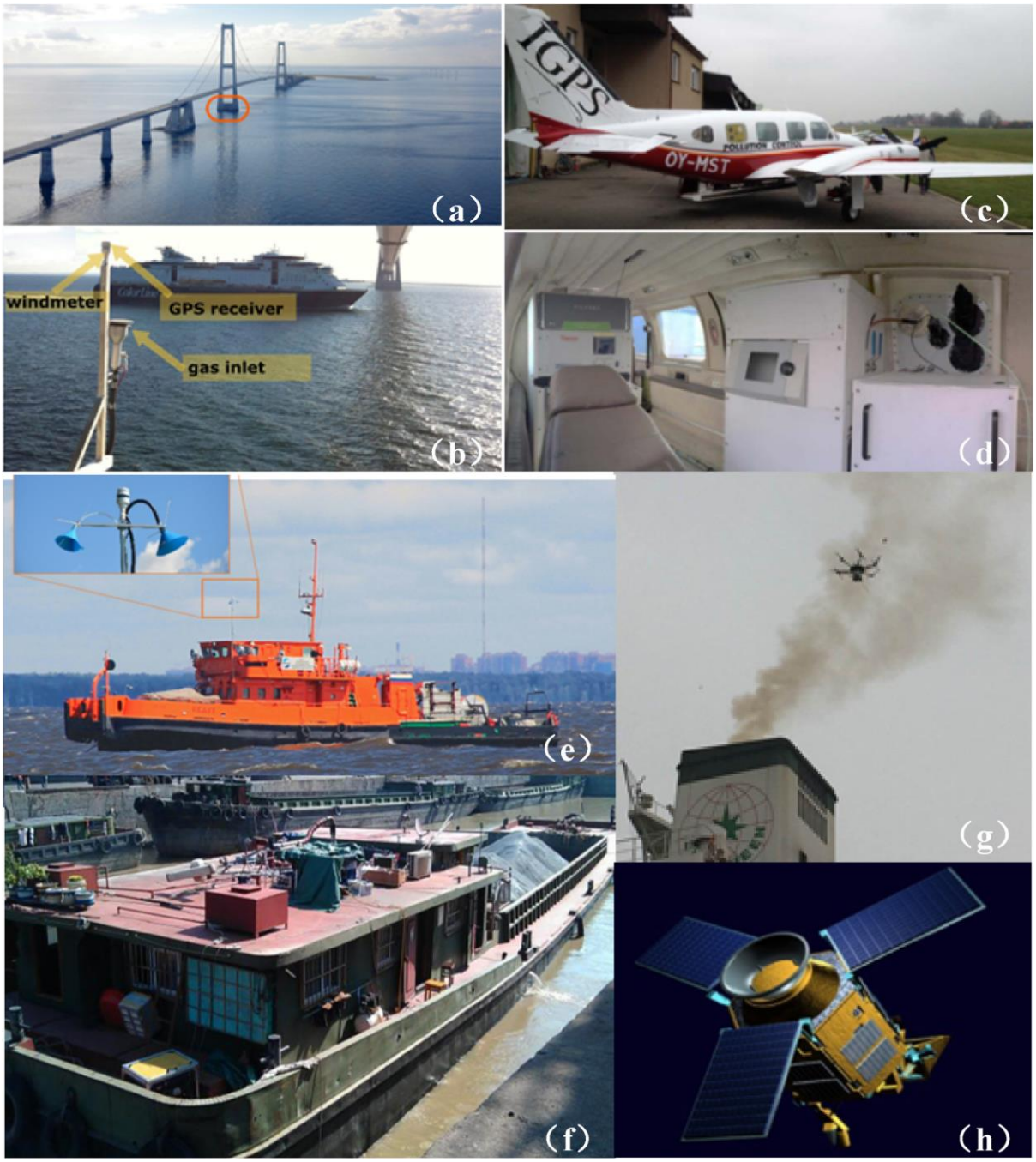
I. Multi-platform collaborative monitoring –

To monitor ship emissions, pollutants such as SO2, CO2, NO, NO2, PM2.5, and PM10, discharged by ships, are considered as the main monitoring object.
Ship emission monitoring methods can be divided into land-based, airborne-based, patrol-boat based, ship-based, satellite-based and unmanned aerial vehicle (UAV)-based method.

a. Land-based – Land-based methods usually deploy equipment along the shore, bridge, or dock, whereby the smoke plume can be measured as a ship passes. The advantage of this approach is that it can provide continuous monitoring; however, it is a passive monitoring method that is greatly affected by wind speed, wind direction, and the distance between the ship and equipment.
b. Airborne-based – Airborne-based methods can effectively monitor sailing ships in ECAs; however, the high operating costs are unsuitable for daily monitoring needs of maritime departments. The ship-based method involves installing equipment on a ship to monitor the real-time emission during the voyage. This method achieves the highest accuracy and precision, but is costly and inefficient, measuring only one ship at a time.
c. Patrol boat-based – This approach is similar to the land-based approach but is more flexible.
d. Satellite-based – is suitable for monitoring large-scale ship emissions but can only monitor NOx and PM emissions from ships, meaning SO2 emissions need to be monitored via a different approach.
e. UAV-based – UAV-based method is flexible, convenient, and relatively inexpensive; however, it can only use small devices because of the corresponding payload.

These methods have their own advantages and scope of application. A single monitoring platform cannot fully monitor ship emissions in ECAs. Collaborative monitoring of multiple platforms is necessary to obtain information on ship emissions at different spatial and temporal scales, which can then provide a deeper understanding of ship emissions. On one hand, the macro-monitoring of ECAs and micro-monitoring of individual ships need to be compared and verified. For example, whole ECAs can be monitored by satellite, individual ships can be monitored by other platforms, and the credibility of macro-monitoring in the ECAs can be demonstrated based on sampling statistics. On the other hand, multiple platforms can track and monitor suspicious target ships. For example, this could be done by land-based monitoring of passing ships with

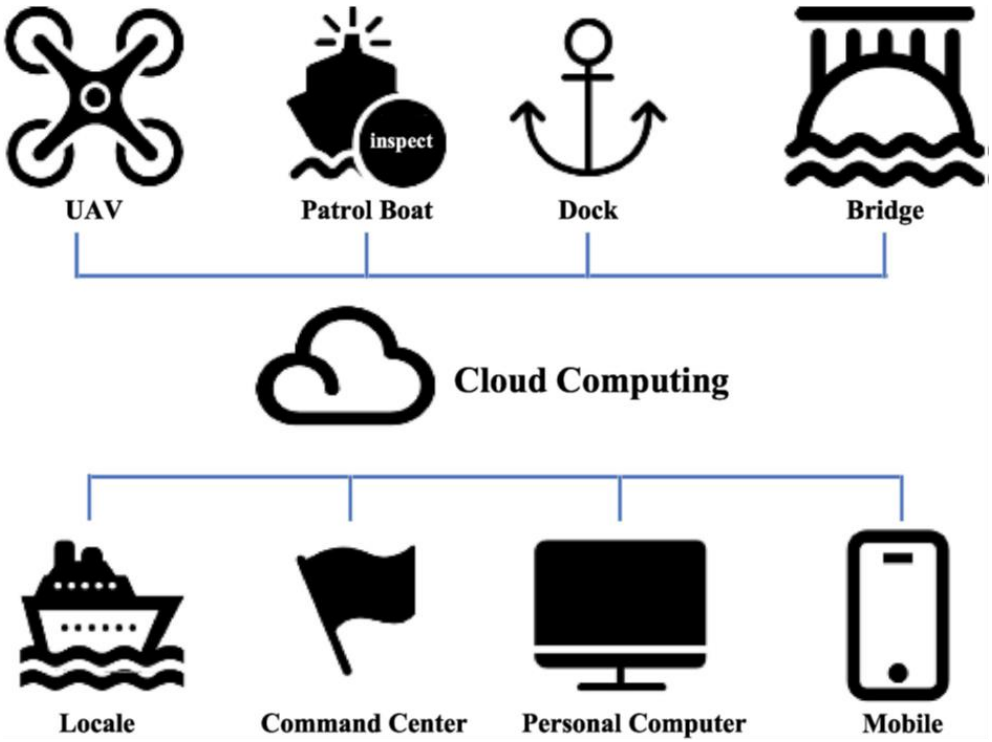
high emissions, and then using UAVs for further tracking, monitoring, and confirmation. In this process, the AIS data provides spatiotemporal information on target ships, providing the necessary data to assess target ships and their emissions. In general, the SEMSW requires a certain collaborative mechanism to monitor ship emissions.



Different ship emission monitoring platforms. (a) and (b) Installation location of land-based equipment and a collector (c) and (d) small aircraft and monitoring equipment
(e) monitoring equipment installed on a patrol boat, which can monitor passing ships
(f) ship-based equipment (g) an unmanned aerial vehicle (UAV) flying into a ship's plume for monitoring purposes.

II. Real-time data transmission and online-processing

The monitoring platforms involved in the proposed SEMSW are diverse and heterogeneous. A gateway model between the sensors and the WWW is included and is responsible for transforming the sensor and its data from the physical world to the WWW. Subsequently, the data collected by the sensors are transmitted to the cloud server centre through wireless communication. Then, online processing converts these data into ship emission information. In this process, the characteristics of ship emissions can be extracted, and an identification model of the emission factors can be established. Finally, the SEMSW aims to provide mechanisms for discovering datasets, defining integrated views of them, continuously receiving data in real-time, visualizing them on screen, and providing a data interaction interface.



Main user interface and visualization interface of the model SEMSW